Demonstration of Proposed Features

Date	15 July 2026
Team ID	XXXXXXXX
Project Name	Rising Waters
Maximum Marks	1 Mark

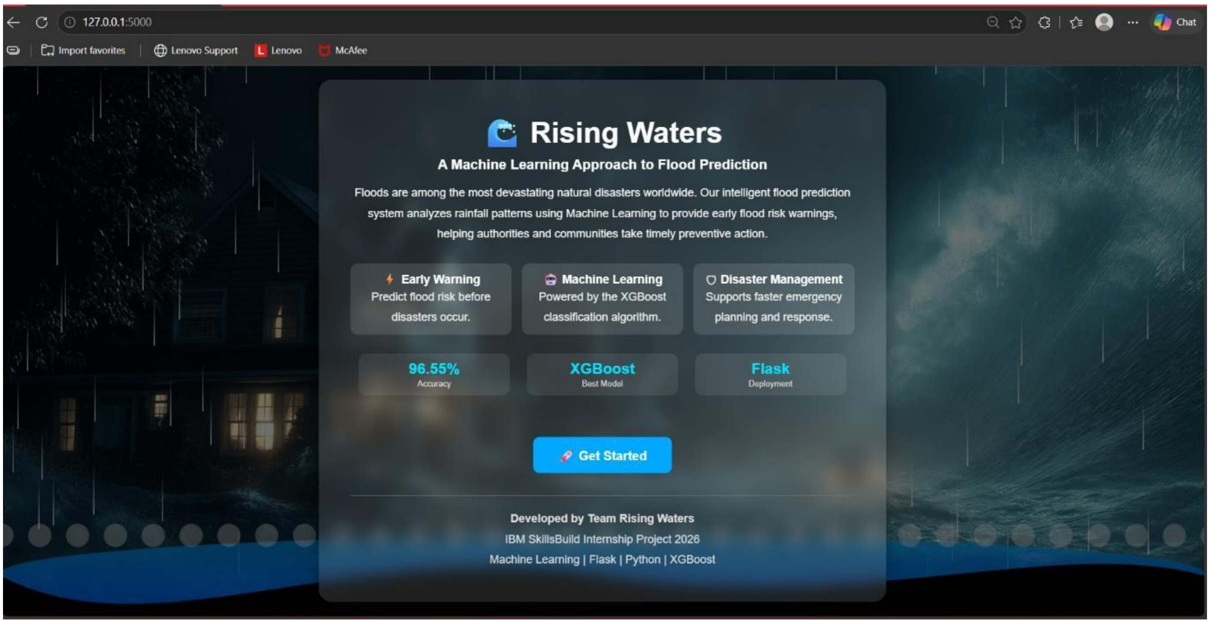
The Rising Waters: A Machine Learning Approach to Flood Prediction system was successfully developed and demonstrated as a web-based application. The application allows users to enter rainfall-related parameters through an interactive interface and instantly predicts whether there is a Flood Chance or No Flood Chance using a trained Machine Learning model

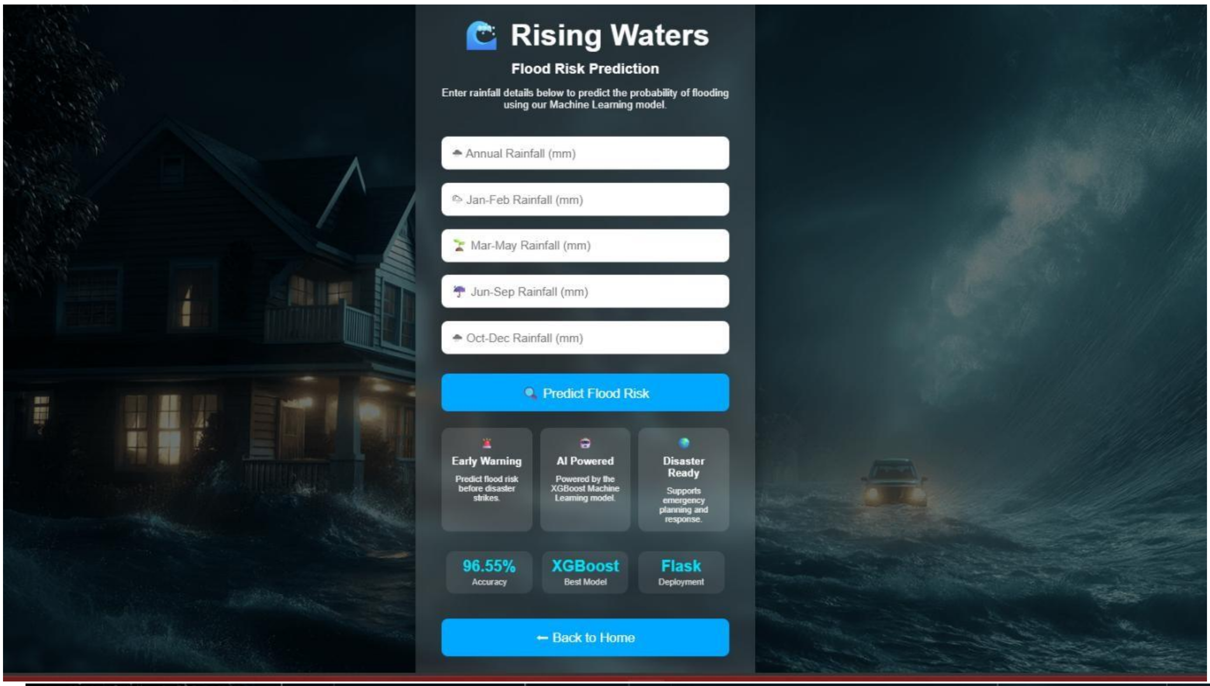
S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Rainfall Data Input	Allows users to enter annual and seasonal rainfall values through a web form.	Implemented	Yes	Successfully accepts user input.
2	Flood Prediction using Machine Learning	Predicts flood occurrence using the trained Random Forest model.	Implemented	Yes	Generates accurate Flood/No Flood predictions.
3	Interactive Web Interface	Responsive frontend developed using HTML, CSS, and JavaScript with rainfall animation.	Implemented	Yes	User friendly interface with Smooth navigation
4	Prediction Result Display	Displays Flood or No Flood result dynamically based on model prediction.	Implemented	Yes	Results are displayed instantly after prediction.
5	Cloud Deployment	Flask application deployed successfully on the Render cloud platform.	Implemented	Yes	Application is publicly accessible through the deployed URL.

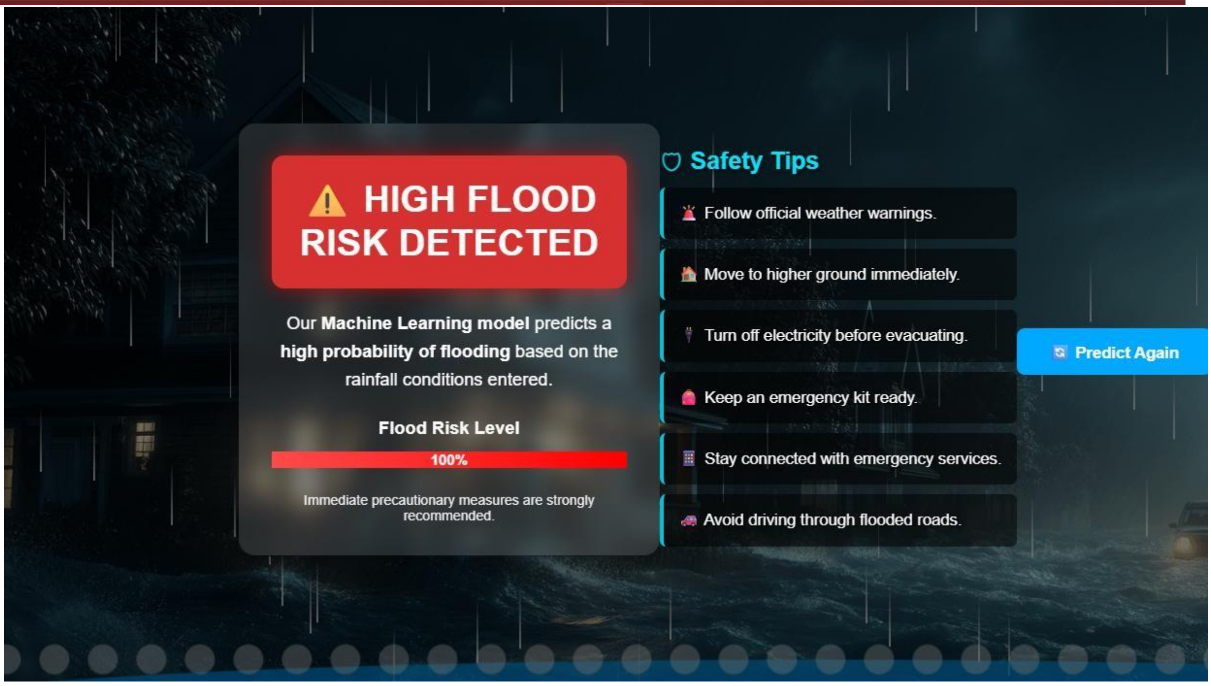
Feature Implementation Summary:

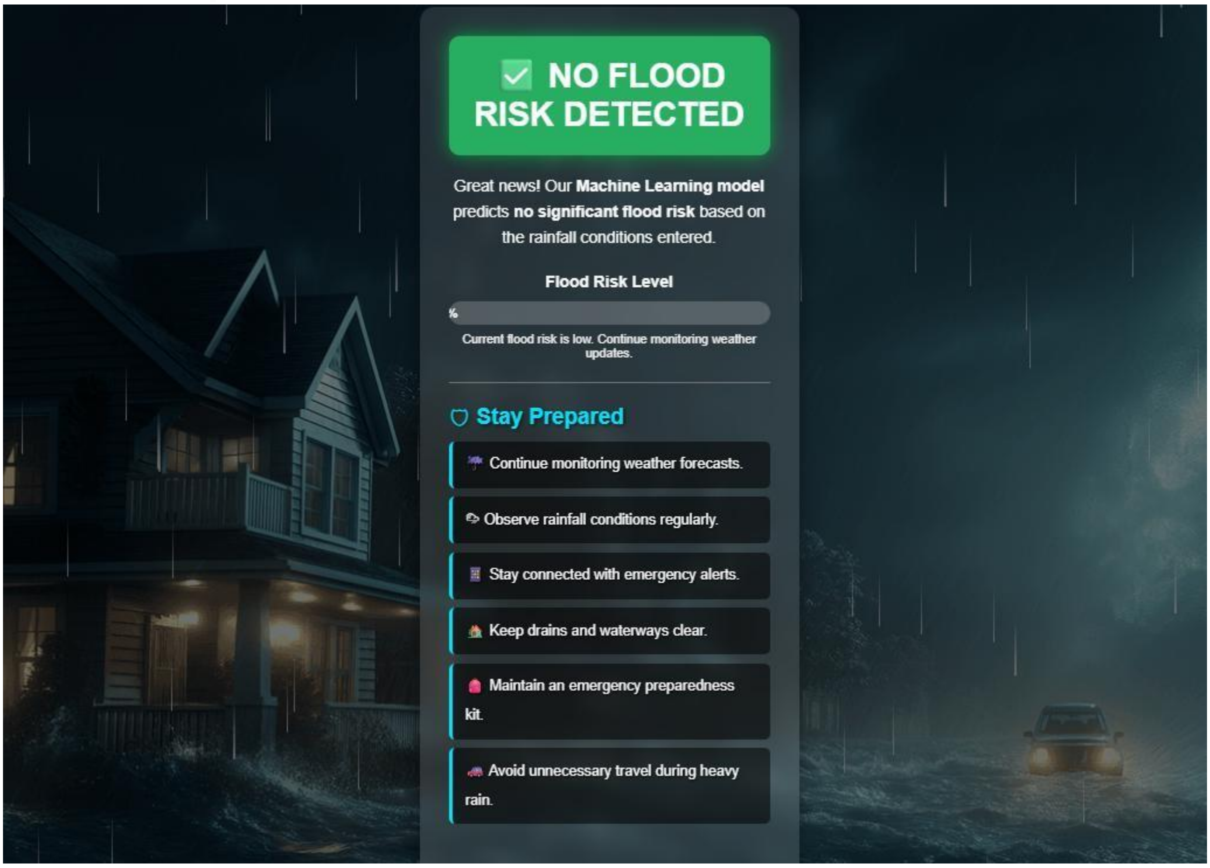
Item	Value
Total Features Proposed	5
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	100%

Attachments:




Rising Waters
Flood Risk Prediction
Enter rainfall details below to predict the probability of flooding using our Machine Learning model.
Annual Rainfall (mm)
Jan-Feb Rainfall (mm)
Mar-May Rainfall (mm)
Jun-Sep Rainfall (mm)
Oct-Dec Rainfall (mm)
Predict Flood Risk
Early Warning
Predict flood risk before disaster strikes.
AI Powered
Powered by the XGBoost Machine Learning model.
Disaster Ready
Supports emergency planning and response.
96.55%
Accuracy
XGBoost
Best Model
Flask
Deployment
Back to Home


HIGH FLOOD RISK DETECTED
Our Machine Learning model predicts a high probability of flooding based on the rainfall conditions entered.
Flood Risk Level
100%
Immediate precautionary measures are strongly recommended.
Safety Tips
Follow official weather warnings.
Move to higher ground immediately.
Turn off electricity before evacuating.
Keep an emergency kit ready.
Stay connected with emergency services.
Avoid driving through flooded roads.
Predict Again


NO FLOOD RISK DETECTED
Great news! Our Machine Learning model predicts no significant flood risk based on the rainfall conditions entered.
Flood Risk Level
%
Current flood risk is low. Continue monitoring weather updates.
Stay Prepared
Continue monitoring weather forecasts.
Observe rainfall conditions regularly.
Stay connected with emergency alerts.
Keep drains and waterways clear.
Maintain an emergency preparedness kit.
Avoid unnecessary travel during heavy rain.